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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Experiment – 7: Database as a Service(DBaaS)

Aim: To create and Connect to a MySQL Database with Amazon RDS. (Free Tier Cloud Platform).

Objective: After performing the experiment, the students will be able to learn –
- Create an environment to run MySQL database
- Connect to the database
- How To Access RDS Database
- Delete the database instance

Lab objective mapped : ITDO8024.3 : To learn and examine cloud service model and deploy database, storage services over different cloud platforms.

Prerequisite:
An AWS account,
MySQL Workbench application.

Requirements: Cloud Login, Desktop, Browser, Internet, GitHub etc.

Pre-Experiment Theory: (describe following topics in soft copy form)

1. **Database services-**

Sr No	**Database type**		**AWS service**
1	Relational Database	Relational databases store data with predefined schemas and relationships between them. These databases are designed to support ACID transactions, and maintain referential integrity and strong data consistency.	1. Amazon Aurora 2. Amazon RDS 3. Amazon Redshift
2	In-memory Database	In-memory databases are used for applications that require real-time access to data. By storing data directly in memory, these databases deliver microsecond latency to applications for whom millisecond latency is not enough.	1. Amazon ElastiCache 2. Amazon MemoryDB for Redis
3	Document Database	A document database is designed to store semistructured data as JSON-like documents. These databases help developers build and update	Amazon DocumentDB (with MongoDB compatibility)

		applications quickly.	
4	Key-value Database	Key-value databases are optimized for common access patterns, typically to store and retrieve large volumes of data. These databases deliver quick response times, even in extreme volumes of concurrent requests.	Amazon DynamoDB

2. **Amazon RDS**
   Amazon RDS is an easy to manage relational database service optimized for total cost of ownership. It is simple to set up, operate, and scale with demand. Amazon RDS automates the undifferentiated database management tasks, such as provisioning, configuring, backups, and patching. Amazon RDS enables customers to create a new database in minutes, and offers flexibility to customize databases to meet their needs across 8 engines and 2 deployment options. Customers can optimize performance with features, like Multi-AZ with two readable standbys, Optimized Writes and Reads, and AWS Graviton3-based instances, and choose from multiple pricing options to effectively manage costs.

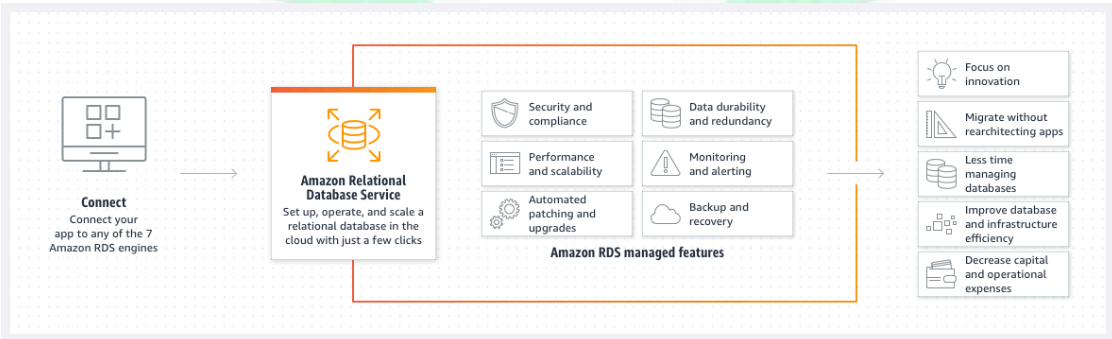


Fig. Amazon Relational Database Service (Amazon RDS) working

Procedure

Step1: Create MY SQL DB instance

1. select AWS RDS service
2. change to N.vergina region
3. select option ‘create database’
4. select option ‘standard create’ and not ‘easy create’
5. select engine option as MYSQL
   edition- MYSQL community
   engine version -MYSQL (some version number)
   Template: free tier
   availability and durability - all disable
6. DB instance configuration steps:
   settings:
   DB instance identifier - Name DB
   Master uname:- admin
   Master pwd: (give some pwd)
   confirm pwd: (retype)
7. Instance configuration:
   DB instance class: Burstable classes
   select db.t2.micro
8. Instance specifications:

storage type: General purpose (SSD)
allocated storage- 20 GB
enable storage autoscaling - disable

9. connectivity
compute resource: -Don't connect to an EC2 compute resource
VPC -default
subnet - default
10. Public accessibility: yes RDS assignes a public IP address.
(This setting will enable direct connection to the database from your device)
11. VPC security group:
create new VPC security group (this will allow connection from the IP address of the device to created database)
Name new security group
12. availability zone: No preference
13. RDS proxy - unchecked
14. Additional configuration:
Port -3306 (default)
15. Database Authentication: select pwd authentication
16. Monitoring:
Enhanced monitoring unchecked [paid service- allow to check metrics in real time for the os that DB instance runs on]
17. Additional Configuration:
Database options
name- dbname(compulsary) AWS RDS does not create it automatically if not specified.
DB parameter group - default
option group - default
Encryption -not for free
Back up - retention period 1 day
Back up window- no preference
copy tag to snapshot - checked
log exports - all unchecked
IAM role -default
Maintainance:
enable auto mirror - checked
maintenance window - no preference
Deletion protection: enable -unchecked
18. select Create Database
DB is being created.
Status changing to Creating to backing-up to Available (indicating DB is running)

Step2: Download SQL workbench client
Download MYSQL Workbench client and install . (No sign-up required)

Step3: Connect to MySQL DB using SQL workbench client
Launch MySQLWorkbench application.
Select Database - connect to database
Enter the following:

Hostname : endpoint info of RDS
Port : 3306
Username and pwd: Use the ones set earlier while creating AWS RDS DB.
Pwd stored in vault - ok

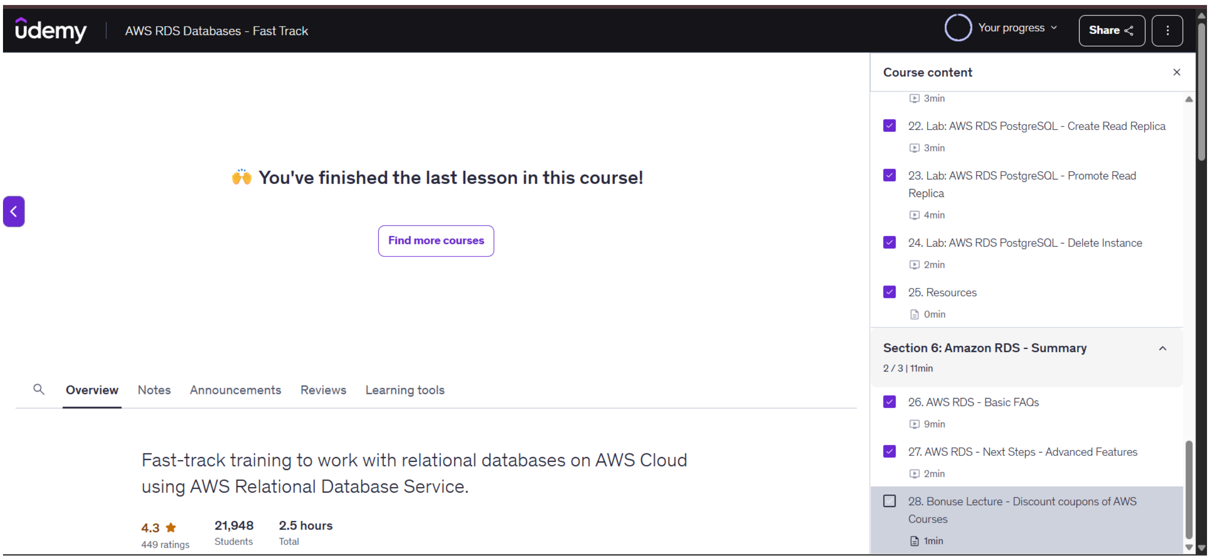
Run the following SQL quires on SQL workbench client
To execute the SQL statement highlight the statement from the coding window and execute. Observe the action output.

```
CREATE  DATABASE ex;
USE ex;
CREATE TABLE st(
  table_id INT PRIMARY KEY,
  stud_id  INT,
  stud_name CHAR(10));
desc st;

INSERT INTO st( table_id, stud_id, Stud_name) values (01, 100, xyz)

SELECT * FROM st;

DELETE FROM st WHERE table=01;

DROP TABLE st;
```

Step 4: Clean up DB instance
Delete the DB instance and security group created inside AWS RDS.

Post-Experiments Exercise

AWS RDS Databases - Fast Track (free course on Udemy):
https://www.udemy.com/course/aws-rds-databases-tutorial-training-free/
complete the course and attach completion proof (like enrollemnet proof, prepare one google_quiz_10mrks_attach link here, proof of notes taken etc)



Extended Theory:

1. Why are databases essential?(to be written in hand)
2. How many DB instances can I run with Amazon RDS?(to be written in hand)
3. Types of AWS Databases(Complete given table)(soft copy)

Database Providers	AWS	AZURE	GOOGLE
NAMES	1. Amazon Aurora 2. Amazon Redshift 3. Amazon RDS 4.	1.	1.

Results/Calculations/Observations:
Fill the following observation tables-(to be written in hand)
Amazon Aurora Vs RDS

Sr. No	**Parameter**	**Amazon Aurora**	**Amazon RDS**
1.	Performance		
2.	Compatibility with Database Engines		
3.	Limitations		

Post Experimental Exercise-(soft copy)
Questions:

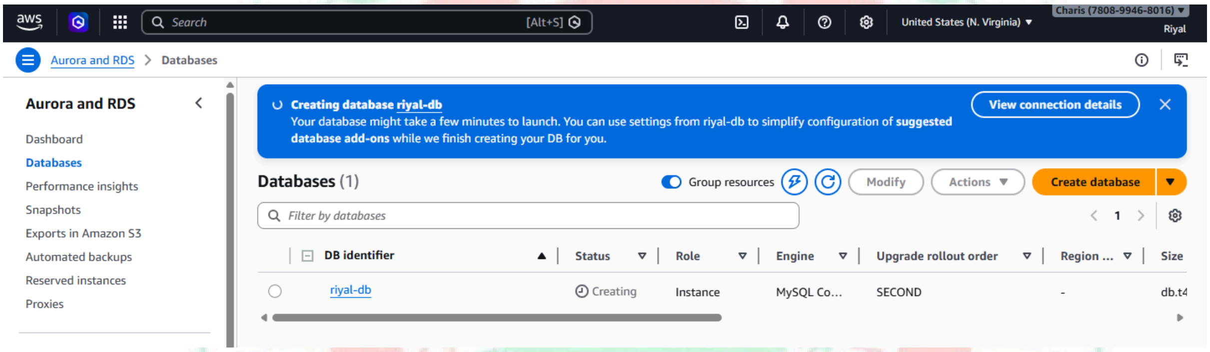
1. What is a database instance (DB instance)?
Ans:A database instance (DB instance) is an isolated database environment in a cloud service like Amazon RDS where a database is created and managed. It includes the compute resources, storage, and database engine required to run and operate the database. Each DB instance acts as an independent database server, allowing users to store, manage, and retrieve data. It can be configured with different performance levels, storage capacities, and security settings based on application needs. Essentially, a DB instance provides a ready-to-use database system without requiring manual setup or maintenance of the underlying infrastructure.

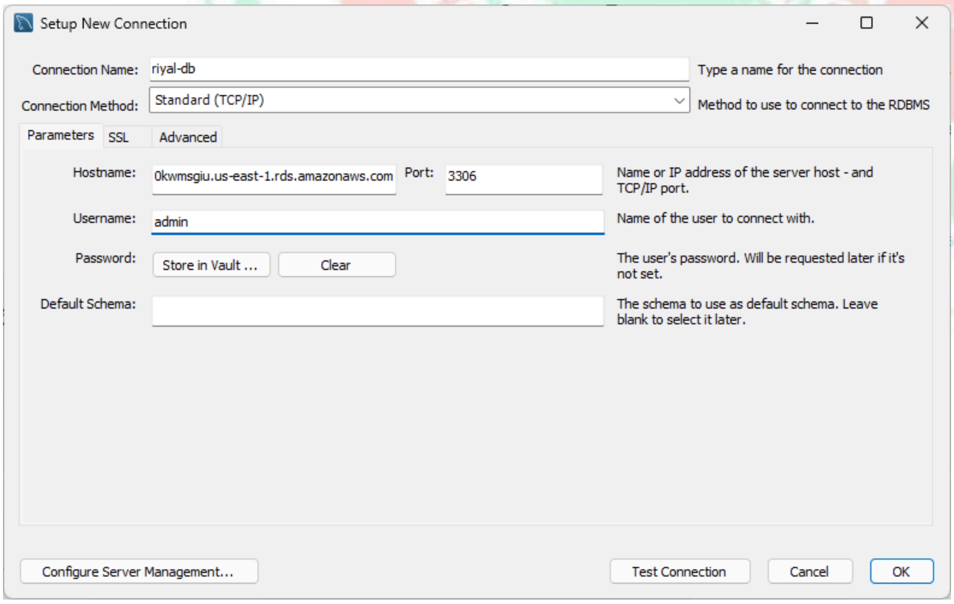
2. How will I be charged and billed for my use of Amazon RDS?
Ans:Amazon RDS follows a pay-as-you-go pricing model, meaning you are charged based on the resources you use rather than a fixed cost. Billing typically includes charges for DB instance usage (based on instance type and running time), storage consumed, backup storage, and data transfer. Additional costs may apply for features like Multi-AZ deployments or read replicas. Charges are calculated hourly or per second depending on the configuration, and the total usage is included in your AWS monthly bill. This flexible billing system allows users to scale their database usage up or down while only paying for what they actually consume.

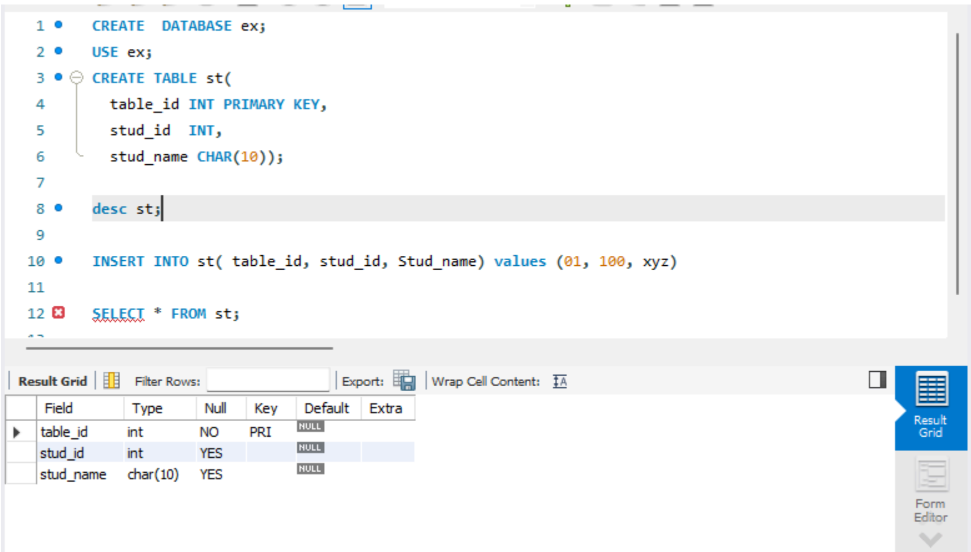
Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

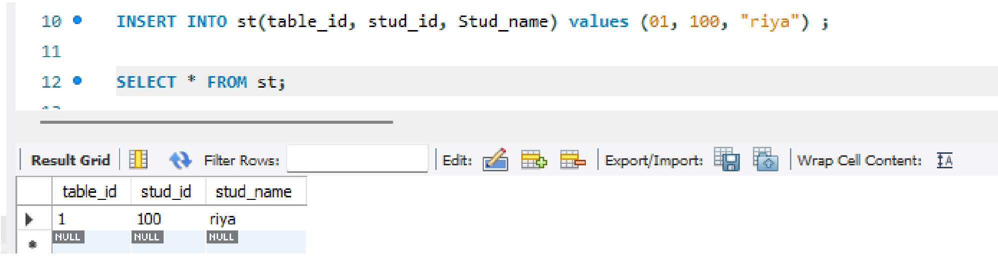
References:

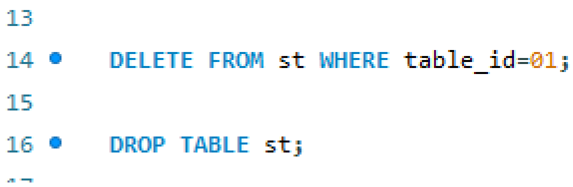
[1]
[Online]https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.RDSMySQL.Import.html#AuroraMySQL.Migrating.RDSMySQL.Space
[2] [Online]https://docs.aws.amazon.com/Am azonRDS/latest/UserGuide/Welcome.html

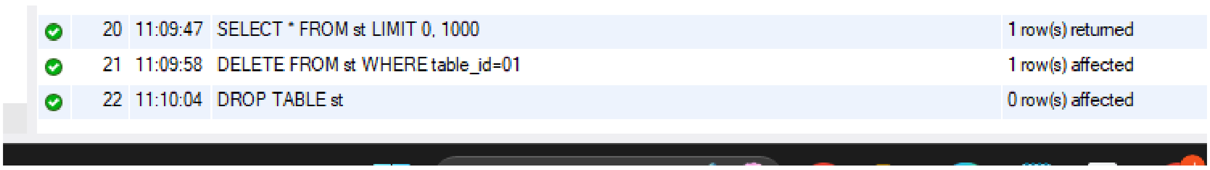


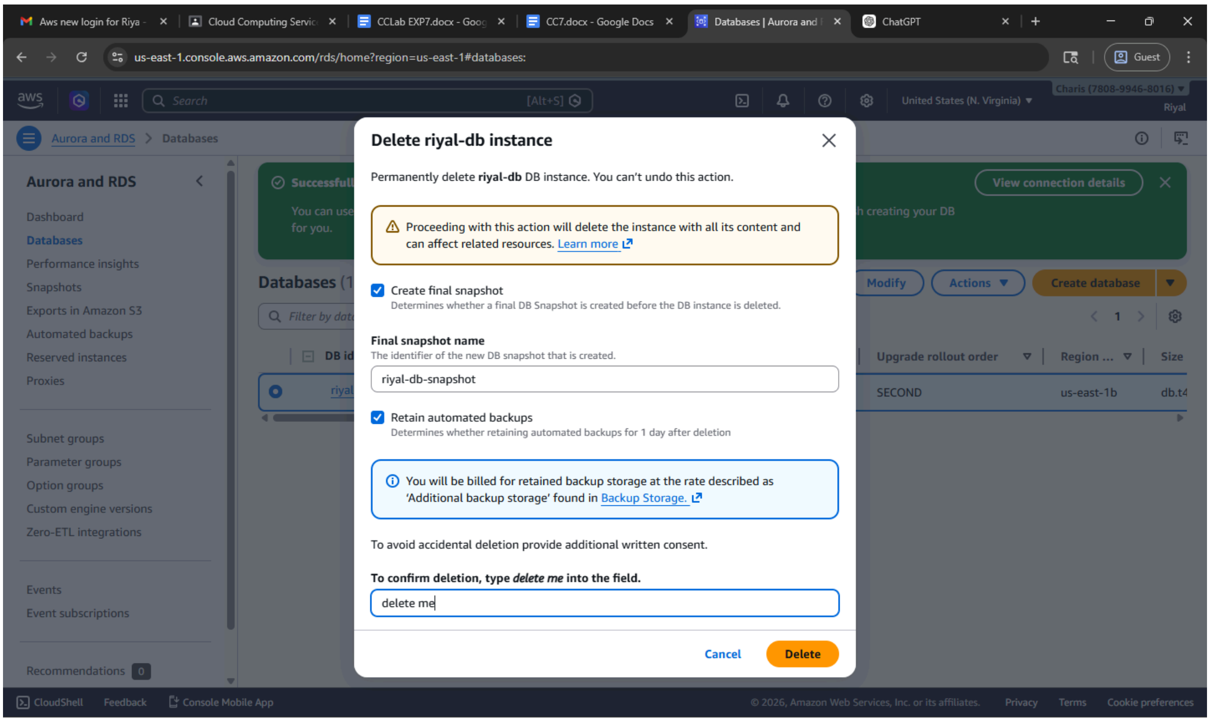





10 INSERT INTO st(table_id, stud_id, Stud_name) values (01, 100, "riya") ;
11
12 SELECT * FROM st;
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
table_id stud_id stud_name
1 100 riya
NULL NULL NULL


13
14 DELETE FROM st WHERE table_id=01;
15
16 DROP TABLE st;


20 11:09:47 SELECT * FROM st LIMIT 0, 1000 1 row(s) returned
21 11:09:58 DELETE FROM st WHERE table_id=01 1 row(s) affected
22 11:10:04 DROP TABLE st 0 row(s) affected


us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#databases:
Aurora and RDS > Databases
Aurora and RDS
Dashboard
Databases
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Event subscriptions
Recommendations 0
Delete riyal-db instance
Permanently delete riyal-db DB instance. You can't undo this action.
Proceeding with this action will delete the instance with all its content and can affect related resources. Learn more
Create final snapshot
Determines whether a final DB Snapshot is created before the DB instance is deleted.
Final snapshot name
The identifier of the new DB snapshot that is created.
riyal-db-snapshot
Retain automated backups
Determines whether retaining automated backups for 1 day after deletion
You will be billed for retained backup storage at the rate described as 'Additional backup storage' found in Backup Storage.
To avoid accidental deletion provide additional written consent.
To confirm deletion, type delete me into the field.
delete me
Cancel
Delete
View connection details
Modify
Actions
Create database
Upgrade rollout order
Region ...
Size
SECOND
us-east-1b
CloudShell
Feedback
Console Mobile App
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